In [1]:
```
!pip install pytrends pandas matplotlib
```

```
Collecting pytrends
  Downloading pytrends-4.9.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: pandas in c:\users\user\downloads\anacon\lib\site-pac
kages (2.3.3)
Requirement already satisfied: matplotlib in c:\users\user\downloads\anacon\lib\site
-packages (3.10.6)
Requirement already satisfied: requests>=2.0 in c:\users\user\downloads\anacon\lib\s
ite-packages (from pytrends) (2.32.5)
Requirement already satisfied: lxml in c:\users\user\downloads\anacon\lib\site-packa
ges (from pytrends) (5.3.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\user\downloads\anacon\lib\s
ite-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\user\downloads\ana
con\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\user\downloads\anacon\lib\si
te-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\user\downloads\anacon\lib
\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\user\downloads\anacon\li
b\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\user\downloads\anacon\lib\si
te-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\user\downloads\anacon\l
ib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\user\downloads\anacon\l
ib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\user\downloads\anacon\lib
\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\user\downloads\anacon\lib\site-
packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\user\downloads\anacon\li
b\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in c:\users\user\downloads\anacon\lib\site-p
ackages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\user\downloads\a
nacon\lib\site-packages (from requests>=2.0->pytrends) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\user\downloads\anacon\lib\si
te-packages (from requests>=2.0->pytrends) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\user\downloads\anacon
\lib\site-packages (from requests>=2.0->pytrends) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\user\downloads\anacon
\lib\site-packages (from requests>=2.0->pytrends) (2025.11.12)
Downloading pytrends-4.9.2-py3-none-any.whl (15 kB)
Installing collected packages: pytrends
Successfully installed pytrends-4.9.2
```

In [3]:
```
# 1. IMPORT LIBRARIES# ===

import pandas as pd
from pytrends.request import TrendReq
import matplotlib.pyplot as plt
from datetime import datetime, timedelta
```

```python
# =====# 2. CONNECT TO GOOGLE TRENDS# ====

print("\nConnecting to Google Trends...")
pytrends = TrendReq(hl='en-US', tz=360)
print("Connection Established Successfully!\n")


# ===# 3. BUILD PAYLOAD# ======

keyword = input("Enter Keyword to Analyze: ")
kw_list = [keyword]

pytrends.build_payload(
    kw_list,
    cat=0,
    timeframe='today 12-m',
    geo='',
    gprop=''
)


# =====# 4. HISTORICAL SEARCH TRENDS#    interest_over_time()# ====

print("\nFetching Historical Search Trends...")

historical_data = pytrends.interest_over_time()

if not historical_data.empty:

    if 'isPartial' in historical_data.columns:
        historical_data = historical_data.drop(columns=['isPartial'])

    print("\nTop 5 Peak Search Dates:")
    print(historical_data.sort_values(by=keyword, ascending=False).head())

    historical_data.to_csv("historical_search_trends.csv")

    plt.figure(figsize=(12,6))
    plt.plot(historical_data.index, historical_data[keyword])
    plt.title(f"Historical Search Trends for '{keyword}' (Last 12 Months)")
    plt.xlabel("Date")
    plt.ylabel("Search Interest (0-100)")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

else:
    print("No Historical Data Found.")


# ==========# 5. HOURLY PRECISION ANALYSIS#    get_historical_interest()# This sect

# print("\nFetching Hourly Precision Data (Last 7 Days)...")

# end_date = datetime.today()
```

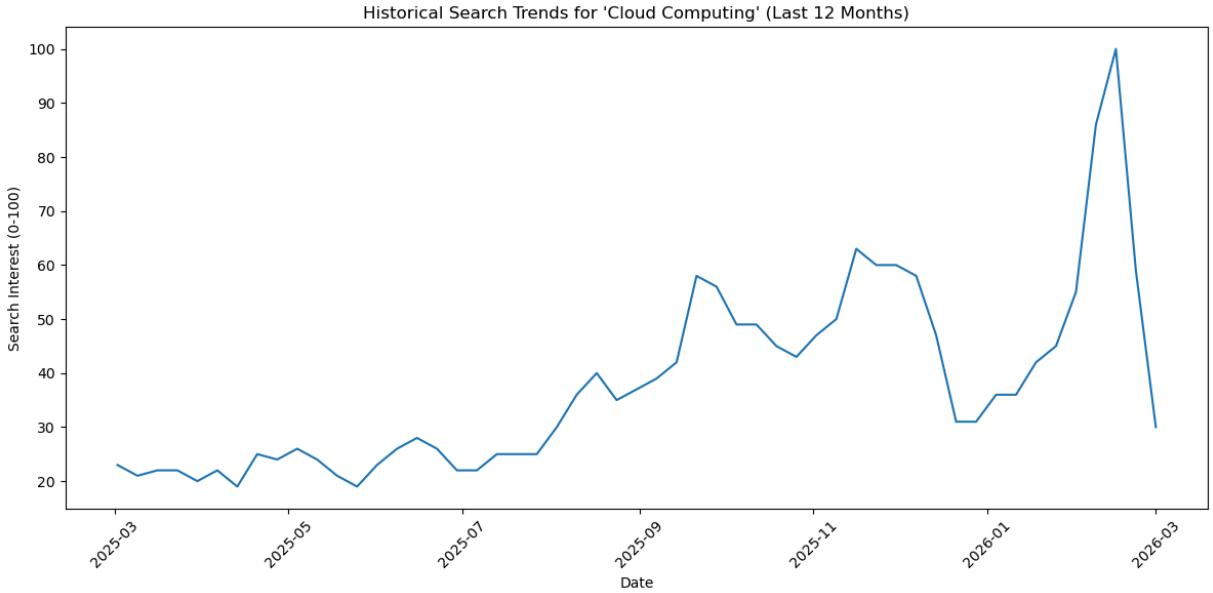
```
# start_date = end_date - timedelta(days=7)

# hourly_data = pytrends.get_historical_interest(
#     kw_list,
#     year_start=start_date.year,
#     month_start=start_date.month,
#     day_start=start_date.day,
#     hour_start=0,
#     year_end=end_date.year,
#     month_end=end_date.month,
#     day_end=end_date.day,
#     hour_end=23,
#     cat=0,
#     geo='',
#     gprop=''
# )

# if not hourly_data.empty:

#     if 'isPartial' in hourly_data.columns:
#         hourly_data = hourly_data.drop(columns=['isPartial'])

#     print("\nTop 5 Peak Hours:")
#     print(hourly_data.sort_values(by=keyword, ascending=False).head())

#     hourly_data.to_csv("hourly_precision_trends.csv")

#     plt.figure(figsize=(12,6))
#     plt.plot(hourly_data.index, hourly_data[keyword])
#     plt.title(f"Hourly Search Trend for '{keyword}' (Last 7 Days)")
#     plt.xlabel("Date & Hour")
#     plt.ylabel("Search Interest (0-100)")
#     plt.xticks(rotation=45)
#     plt.tight_layout()
#     plt.show()

# else:
#     print("No Hourly Data Found.")


# ======== 6. REGIONAL INTEREST ANALYSIS#    interest_by_region()# ========

print("\nFetching Regional Interest Data...")

regional_data = pytrends.interest_by_region()

if not regional_data.empty:

    regional_data = regional_data.sort_values(by=keyword, ascending=False)

    print("\nTop 10 Regions:")
    print(regional_data.head(10))

    regional_data.to_csv("regional_interest.csv")

    top_regions = regional_data.head(10)
```

```
    plt.figure(figsize=(12,6))
    plt.bar(top_regions.index, top_regions[keyword])
    plt.title(f"Top 10 Regions Searching '{keyword}'")
    plt.xlabel("Region")
    plt.ylabel("Search Interest (0-100)")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

else:
    print("No Regional Data Found.")


# ============== 7. RELATED QUERIES#    related_queries()# ======

print("\nFetching Related Queries...")

related_queries = pytrends.related_queries()

if keyword in related_queries:

    print("\nTop Related Queries:")
    print(related_queries[keyword]['top'])

    print("\nRising Related Queries:")
    print(related_queries[keyword]['rising'])


# ======= 8. KEYWORD SUGGESTIONS#    suggestions()# ========

print("\nFetching Keyword Suggestions...")

suggestions = pytrends.suggestions(keyword=keyword)
suggestions_df = pd.DataFrame(suggestions)

print("\nKeyword Suggestions:")
print(suggestions_df)

suggestions_df.to_csv("keyword_suggestions.csv", index=False)


# =========# PROJECT COMPLETED# ===========

print("\n==================================================")
print(" GOOGLE SEARCH ANALYSIS COMPLETED SUCCESSFULLY ")
print(" CSV FILES SAVED IN PROJECT FOLDER             ")
print("==================================================")
```

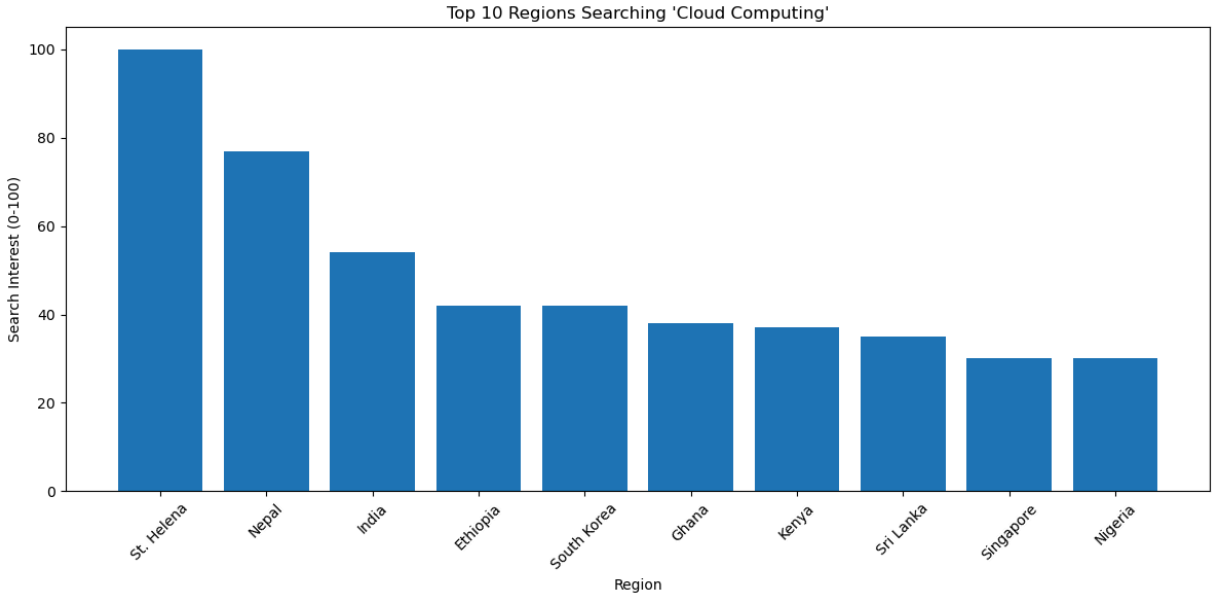
```
Connecting to Google Trends...
Connection Established Successfully!

Fetching Historical Search Trends...
```

```
C:\Users\user\Downloads\anacon\Lib\site-packages\pytrends\request.py:260: FutureWarn
ing: Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated and wi
ll change in a future version. Call result.infer_objects(copy=False) instead. To opt
-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)
`
  df = df.fillna(False)
```

```
Top 5 Peak Search Dates:
            Cloud Computing
date
2026-02-15              100
2026-02-08               86
2025-11-16               63
2025-11-30               60
2025-11-23               60
```




```
Fetching Regional Interest Data...

Top 10 Regions:
             Cloud Computing
geoName
St. Helena               100
Nepal                     77
India                     54
Ethiopia                  42
South Korea               42
Ghana                     38
Kenya                     37
Sri Lanka                 35
Singapore                 30
Nigeria                   30
```


Top 10 Regions Searching 'Cloud Computing'
Search Interest (0-100)
0
20
40
60
80
100
St. Helena
Nepal
India
Ethiopia
South Korea
Ghana
Kenya
Sri Lanka
Singapore
Nigeria
Region

```
Fetching Related Queries...

Top Related Queries:
                          query  value
0          cloud computing news    100
1       what is cloud computing     96
2      cloud computing services     67
3        google cloud computing     37
4   what is the cloud computing     32
5      cloud computing security     31
6           cloud computing pdf     31
7       cloud computing service     29
8           aws cloud computing     25
9                           aws     24
10         free cloud computing     21
11     cloud computing software     21
12  services of cloud computing     20
13            quantum computing     19
14   cloud computing definition     17
15     cloud computing benefits     17
16  security in cloud computing     16
17   cloud computing technology     16
18      cloud computing meaning     15
19                ai news today     14
20                        azure     14
21       cloud computing course     12
22               edge computing     11
23      cloud computing updates     11
24         cloud computing jobs     11

Rising Related Queries:
                          query   value
0                ai news today  252850
1           markets news today   27550
2     cybersecurity news today   24750
3         cloud computing news    6050
4      cloud computing updates    1500
5            quantum computing     500
6          cloud computing pdf     180
7     cloud computing services     120
8         free cloud computing     100
9       google cloud computing      70
10  cloud computing definition      60
11    cloud computing benefits      50

Fetching Keyword Suggestions...

Keyword Suggestions:
             mid                         title               type
0     /m/02y_9m3               Cloud computing  Organization type
1   /g/12bm3rm_7     Cloud Computing Explained              Topic
2     /m/0jzt0th  Cloud computing architecture              Topic
3     /m/09v8lc9      Cloud computing security              Topic
4  /g/11c0q_754d          Serverless computing              Topic

==============================================
```

```
 GOOGLE SEARCH ANALYSIS COMPLETED SUCCESSFULLY
 CSV FILES SAVED IN PROJECT FOLDER
==============================================
```

In [ ]:

```

```